

Autonomous Navigation & Mapping of Small Bodies

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Abstract—Autonomous mapping and navigation around unknown small bodies is a challenging problem. In today's missions, small body mapping and navigation (SBMN) require significant human intervention on the ground for map refinement and supervision of the navigation and orbit selection process. Although current methodologies adequately performed in past missions (e.g., Rosetta, Hayabusa, Deep Space), they are not suitable for applications requiring a high level of autonomy. This work proposes a method for autonomous orbit selection and adaptation around a small body while mapping its surface. In particular, in this work, we will develop cost functions that quantify the orbit goodness in the sense of map improvement. In other words, we develop quantitative measures that characterize the accuracy of the small body map and use these measures in an optimization process to compute the next best orbit that maximally contributes to the map enhancement. The proposed framework reduces the human involvement in this process and takes a step toward the fully autonomous mapping and navigation around small bodies.

challenge [1]. In this work, we discuss such challenges for missions designed to approach and map the surface of unknown objects (e.g., asteroids or comets). Dealing with such challenging environment with high levels of uncertainty requires human intervention across different phases of the mission. In particular, the current state-of-the-art techniques heavily rely on human intervention in the first part of the proximity operations phase (Figure 1a). Moreover, for an adequate mapping and acquisition of scientific data, proper maneuvers have to be designed. Currently, trained analysts play a crucial role in this process by designing trajectories and assisting in the mapping of small bodies. This human-in-the-loop process is both time consuming and limits the application of the methods to many future missions, e.g., sending multiple spacecraft to explore several small bodies simultaneously, see Figure 1b. The aim of this work is to take a step toward full spacecraft autonomy. In particular, we focus on the orbit selection process to enhance mapping accuracy and coverage of small bodies.

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1. INTRODUCTION

In the last decade, many space agencies and industries have focused on enhancing the automation capabilities of spacecraft. While autonomy is crucial for future spacecraft, adding autonomy to space missions is a chal-

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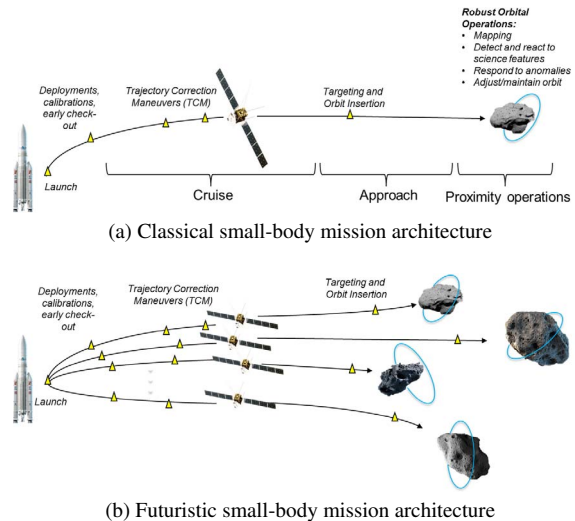


Figure 1. Small-body mission architecture

Following the literature in robotic autonomy, the work in this paper is most related to literature on Active SLAM (simultaneous localization and mapping) [2]. The

general problem is to localize the spacecraft with respect to the small body and plan the best trajectory that would improve the mapping of the small body. In this paper we focus on a reduced version of the active SLAM problem. In particular, we will present a path planning algorithm that maximizes the information gain in mapping accuracy. Abstracting the gravity mapping, the main goal of this paper is to enable the spacecraft to autonomously take actions to improve the knowledge of the environment (i.e., small body shape).

The most common techniques for small body mapping are stereophotogrammetry (SPG) [3] and stereophotoclinometry (SPC) [4]. In this paper, we focus on the SPC approach and aim to build an autonomous orbit selection framework to enhance SPC performance. The main idea of SPC is to build small digital topography and albedo maps (L-map) from images of the same area under different illumination conditions. Once a map is acquired, the rendered visual appearance of the map is predicted and compared to actual images. Then, the center of the map (landmark) and the spacecraft pose can be estimated by iteratively minimizing the difference between expected map appearance and actual image. Existing small body mapping techniques (e.g., [4], [3], [5]) require human supervision and they are typically decoupled from the trajectory design process. Regarding the localization methods for space applications, one of the most common methods is the AutoNav approach [6], which uses a batch filter where data are stored on-board, edited, and then input to the filter. Other example localization systems include landmark-based filtering techniques using the extended Kalman filter [7], higher order additive divided difference sigma points filter [8], unscented Kalman filter [9] and particle filter [10]. However, the problem of joint estimation and autonomous decision making (i.e., active SLAM) is a relatively less explored area in the space literature. Most related work to active SLAM is focused on landing application (e.g., [11]).

On the other hand, active SLAM has received a lot of attention in the robotics community. Feder et al. [2] propose an algorithm to determine the optimal action to maximize the next step information gain given the current knowledge of the map. Bourgault [12] improved [2] by including a global term in the utility function that takes into account, for each possible action, the reduction of the uncertainty of the map represented by an occupancy grid. More recent works have proposed a belief space path planner with minimum uncertainty by utilizing filters and pose graph representations [13], [14], [15].

In this work, we propose a framework for simultaneous mapping and planning around small bodies. Due to the high complexity of the environment and continuous solution space, the exact problem formulation is computationally intractable. In this work, we propose an approach to reduce the problem to a computationally tractable one. In particular, we propose an algorithm that enables the spacecraft to autonomously and adaptively select the best orbit given a reduced set of possible future orbits. We incorporate mapping accuracy as the driver

for the optimization process. We start by deriving formal quantitative measures that characterize the accuracy of the small body map and use these measures in an optimization process to compute the next best orbit that maximally contributes to the map enhancement.

The highlights and contributions of the proposed method can be summarized as (1) developing formal definitions of quantitative measures that characterize the accuracy of the small body map and (2) proposing an autonomous orbit selection method to maximize the small body coverage. The proposed approach takes a step towards enhancing the autonomy capabilities of the spacecraft during proximity operations with a small body. It also reduces the human effort during trajectory planning and mapping phases.

The paper is organized as follows. In Section 2, we describe a representative mission architecture with a focus on the mission scenario and assumptions. In Section 3 the problem of planning under uncertainty is formulated in a principled manner. Section 4 presents a description of the adopted techniques for mapping and localization and casts the small body mapping and spacecraft localization as a POMDP (partially observable Markov decision process) problem. In Section 5, we describe the autonomous planning problem along with the proposed solution. In Section 6, a description of mapping accuracy in terms of coverage score is introduced and discussed, and Section 7 presents the results.

2. MISSION ARCHITECTURE

A mission to a small body can be divided into a few main operational phases based on the distance to the body (see Figure 2). In particular, after the deep space phase, the spacecraft arrives at the body. From previous and future missions (i.e., Rosetta, OSIRIS-REx), the arrival phase starts when the ratio between the distance and the object's maximum dimension (also referred to as the "interest ratio") becomes smaller than 25. The exact value of this threshold depends on the mission which is slightly different for any mission/body. After arrival, an initial characterization phase starts. For this stage, the interest ratio is about 14-18. The main objective of the characterization phase is to identify landmarks on the small body surface and estimate their position to determine the rotation state of the body and construct an initial shape model. For each part of the body surface several images need to be taken to construct an accurate enough map.

Figure 2 shows the operational sequence and phases for the Rosetta mission as an example. In the Rosetta mission, in the characterization phase images were acquired every hour, and during the initial days of the characterization phase, a first set of landmarks was manually identified by the operators. The next phase is the global mapping phase (where the interest ratio is roughly between 4 and 7). At this stage, initial estimates of the gravity potential and rotational state of the body, landmark positions and shape, and topological model are

being enhanced and refined. Finally a close observation orbit (interest ratio is roughly 2 to 3) is needed for the selection of possible landing sites.

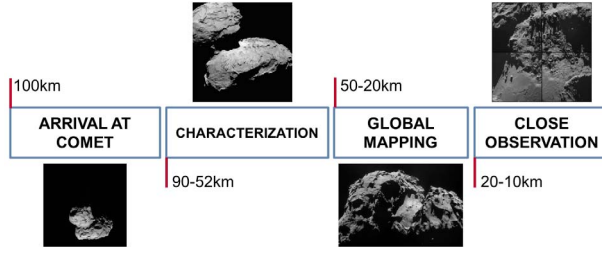


Figure 2. Rosetta operational phases

The relative distance and dimensions of the small body cannot be perfectly known during these phases. However, the projection of the object on the camera plane, measured in pixel, can be a good indicator of the corresponding phase. Figure 3 reports the size of the object maximum dimension on the sensor in pixel in the different phases of the Rosetta mission. A simplified 2D analysis has been carried out. The plot is obtained considering the OSIRIS wide angle camera used in the Rosetta mission and a constant 2D object projection of 4 km in its maximum dimension. The OSIRIS cam specifications are reported in Table 1.

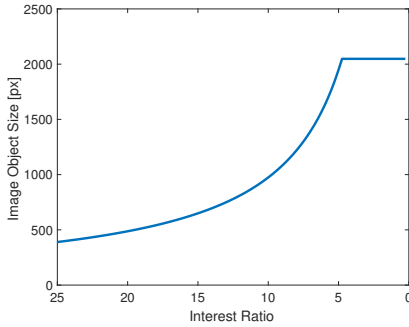


Figure 3. Small body dimension

Mission Scenario And Assumptions

Traditionally, the deep space trajectory and the arrival at the comet are designed on the ground before the launch. However, for future missions, with more complex, uncertain, and far bodies, autonomy is crucial to mission success. In particular, automating the characterization phase can significantly help to reduce the human intervention and enabling complex future missions. Thus,

Table 1. OSIRIS Camera Specifications

Field of view	12°(y) x 12°(x)
Detector size (pixels)	2048 x 2048
Focal length	140 mm

in this paper, we mainly focus on the characterization and global mapping phases. At these phases, there exists a very rough estimate of the body shape and rotational state. This information results from the arrival phase and pre-launch calculations based on light-curve analysis (see Figure 4).

3. PLANNING UNDER UNCERTAINTY

This work is concerned with the problem of autonomous mapping with planning around small bodies. This is an instance of the general problem of planning under motion, sensing, and map uncertainty also known as active SLAM [2]. In its most general form, active SLAM can be cast as a partially observable Markov decision process (POMDP) problem [16] [17] [18].

POMDP is one of the most principled ways of modeling the process of sequential decision making under uncertainty. To describe an abstract POMDP problem, let us denote the system state, action, and observation at the k -th time-step by x_k, u_k, z_k , respectively. Let $x_{k+1} = f(x_k, u_k, w_k)$ and $z_k = h(x_k, v_k)$ denote the system dynamics and measurement models where $w_k \sim p_w(\cdot|x_k)$ and $v_k \sim p_v(\cdot|x_k)$ denote the state-dependent process and observation noise. Due to the observation noise, the best one can infer that the system state is a probability distribution over all possible states, referred to as belief $b_k = p(x_k|z_{0:k})$. Belief space (i.e., the set of all beliefs) is denoted by $\mathbb{B}$. Belief is typically evolved using a recursive filter denoted by τ as $b_{k+1} = \tau(b_k, u_k, z_{k+1})$. The control policy π in a partially observable setting is a mapping from belief space to the action space, i.e., $u_k = \pi(b_k)$. By defining the one-step reward of taking action u at belief b by $\mathcal{R}(b, u)$, we can formally define the T-step POMDP problem as:

$$\pi_{0:T}^* = \arg \max_{\Pi_{0:T}} \mathbb{E} \sum_{k=0}^T \mathcal{R}(b_k, \pi_k(b_k)) \quad (1)$$

Π is the space of all possible policies. Often, in solving POMDPs in order to reduce complexity, the set of admissible policies Π is reduced to a simpler subset over which the optimization can be carried out. In the next section, we describe concrete realization of these terms and the POMDP problem associated with small body mapping. Then, we aim to approximate and solve this problem to compute a policy for spacecraft motion.

4. SMALL BODY MAPPING AND SPACECRAFT LOCALIZATION

In this section, we briefly review the mapping and state estimation around small bodies and discuss how its elements correspond to the abstract POMDP problem described in Section 3.

System Definition

In the small body mapping and planning problem the overall system state x_k is represented by the spacecraft

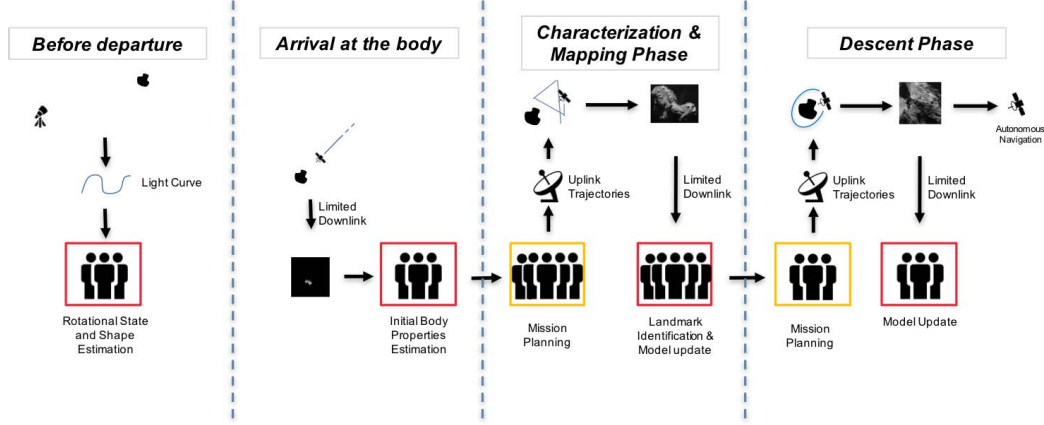


Figure 4. Operations timeline

state and the map state $x_k = (s_k, m_k)$, where $s_k = (\rho_k, \dot{\rho}_k) \in \mathbb{R}^6$ being the spacecraft position and translational velocity relative to the body. m_k represents the state of the map of the body. In this work, we assume that the spacecraft is always pointing towards the center of the small body. Thus, we do not include the spacecraft attitude in the state definition.

Map Representation

In this work, we rely on the SPC technique [4] for mapping. SPC is the main method utilized in small body missions to generate the digital topography/albedo map of the body from multiple images. In SPC, the small body surface map m is represented by a set of maplets with their associated digital topology and albedo. In this paper, we only work with the topology and not albedo. Formally, the map m is then defined as a mesh with n facets, i.e., $m = (f^1, f^2, \dots, f^n)$. The i -th facet $f^i = (v^{i1}, v^{i2}, v^{i3}, \hat{n}^i)$ is a triangle characterized by three vertices v^{i1} , v^{i2} , and v^{i3} . $\hat{n}^i$ denotes the normal vector of the facet, which (is a dependent variable and) can be computed based on the three vertices.

System Belief

Since the spacecraft observations z_k (i.e., images) are noisy, one can only create a probability distribution function (pdf) over all possible system states. We refer to this pdf as the joint belief over the spacecraft state and the map state as denoted by $b^x = p(x|z_{0:k}, u_{0:k})$.

Joint State Estimation

Typically, the joint belief is evolved recursively using a filter. We represent the recursive update of the belief under filter τ as

$$b_{k+1}^x = \tau(b_k^x, u_k, z_{k+1}) \quad (2)$$

where u_k is the control command sent to the spacecraft and z_k is the observation (image) received from spacecraft cameras. For this joint state estimation, we follow the common framework as in space missions. In

particular, we rely on AutoNav [6] for the spacecraft state estimation and on SPC [4] for the map estimation and update. The AutoNav uses a batch filter where data are stored, edited, and then input to the filter. The details of AutoNav and SPC are beyond the scope of this paper. The reader is encouraged to see [6] and [4] for further details.

5. AUTONOMOUS PLANNING FOR SMALL BODY MAPPING (PSBM)

Leveraging small body mapping and spacecraft localization from Section 4, in this section, we discuss the problem of trajectory planning around small bodies for efficient mapping. Planning in an environment that is strongly perturbed, unknown, and subject to very complicated dynamics, as the one around a small body, is noticeably different from the classical deterministic planning problems. A wrong choice of an action, due to unmodeled dynamics or unknown environment, could suddenly cause a collision with the body or generate an escape trajectory. In its most general form the problem is a POMDP (equation 1), whose solution is computationally intractable. To reduce the problem to a computationally tractable one, we start by approximating the policy space Π in equation 1 with a smaller set $\bar{\Pi}$ as described below.

Reduced Policy Space

The primitive spacecraft actions $u \in \mathbb{U}$ are thruster commands; hence the full policy $\pi : \mathbb{B} \rightarrow \mathbb{U}$ is a mapping from the joint belief space to the thruster commands $u = \pi(b)$. However, planning in the continuous space $\mathbb{U}$ of low-level thruster commands is a challenge. Thus, in this work, we first reduced the command space to $\bar{\mathbb{U}}$, which includes a set of desirable orbits to choose from. In other words, instead of selecting a low-level thruster command, we select a target orbit and rely on an existing controller to generate thrust commands that take us to the target orbit. We refer to thruster commands as actions and refer to orbit commands as macro-actions (since each

macro-action is composed of a sequence of low-level actions). In the following, we discuss how each macro-action is parametrized.

Trajectory Definition—a spacecraft trajectory can be defined in several different ways. Following the classical planning formulation, we define the trajectory via a time-stamped sequence of spacecraft state and actions.

$$\mathcal{T} = (s(t), u(t)) \quad (3)$$

Orbit Parameters—an orbit is a certain trajectory that can be parametrized in different ways. We represent the orbit with the following parameters:

$$\omega = (s_0, t_\omega, t_z) \in \Omega \quad (4)$$

where s_0 is the initial state (position and velocity) of the spacecraft at the beginning of the orbit. $t_\omega \in \mathbb{R}_{\geq 0}$ represents how long the spacecraft needs to stay on the orbit ω . Vector $t_z \in \mathbb{R}_{\geq 0}^\eta$ is the sequence of image acquisition time along the orbit. Variable $\eta \in \mathbb{N}$, dictated by the mission constraints, is the upper limit on the number of images that can be taken during time t_ω . Set Ω is the space of all possible ω , which is the reduced action space, i.e., $\bar{\Omega} = \Omega$. Please note that to recover $\mathcal{T}$ from ω it is necessary to rely on a dynamical model of the motion of the spacecraft that also takes into account the gravity field of the small body.

Reduced Policy—consequently, the reduced policy is defined as

$$\bar{\pi} : \mathbb{B} \rightarrow \Omega, \quad \omega = \bar{\pi}(b) \quad (5)$$

$\bar{\Pi}$ as the set of all possible $\bar{\pi}$. To compute the best policy, we first define the reward and value functions.

One-orbit Reward—similar to the one-step reward, we define one-orbit reward $\bar{\mathcal{R}}(b_k, \omega_k)$ for macro-actions. The reward $\bar{\mathcal{R}}$ represents the gain in map accuracy for a given orbit, and is discussed in depth in Section 6.

Value Function—accordingly, value function $\bar{J}$ is defined as the running-sum of all the rewards:

$$\bar{J}(b_0; \bar{\pi}) = \sum_{k=0}^T \bar{\mathcal{R}}_k(b_k, \bar{\pi}(b_k)) = \sum_{k=0}^T \bar{\mathcal{R}}_k(b_k, \omega_k) \quad (6)$$

Reduced Planning Problem—optimizing over the reduced policy space, the planning problem in equation 1 is reduced to finding the optimal $\bar{\pi}^*$ given a defined subset of possible orbits Ω :

$$\bar{\pi}^* = \arg \max_{\bar{\pi}} J(b_0; \bar{\pi}) \quad (7)$$

The block diagram in Figure 5 represents the autonomous planning architecture.

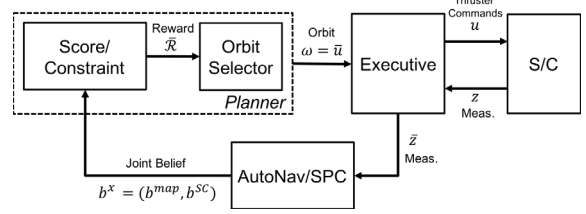


Figure 5. Autonomous planning architecture

6. MAPPING ACCURACY

In this section, we discuss the highlight of this work. For a given orbit, we derive a quantitative measure to capture the information gain in map accuracy. This measure will drive the planning framework. We denote this measure (or reward) by $\bar{\mathcal{R}}$. The reward is a function of the system belief b_k and the selected orbit ω_k , i.e., $\bar{\mathcal{R}}(b_k, \omega_k)$. The system belief b_k encodes the accuracy of map representation as well as the spacecraft state. ω_k is the granularity of the decision variable. The reward over belief can be written as

$$\bar{\mathcal{R}}(b_k, \omega_k) = \mathbb{E} \bar{\mathcal{R}}_x(x_k, \omega_k), \quad x_k \sim b_k(x) \quad (8)$$

where, $\bar{\mathcal{R}}_x(x_k, \omega_k)$ is the information gain measure for a given state of the system (i.e., map and spacecraft) x_k . $\mathbb{E}[\cdot]$ represents the expectation operator.

Computing the exact value of the expectation in equation 8 for a general belief is computationally intractable. The expectation can be approximated via a finite sum or via a most likely state assumption:

$$\bar{\mathcal{R}}(b_k, \omega_k) \approx \sum_{i=1}^n \bar{\mathcal{R}}_x(x_k^{(i)}, \omega_k), \quad x_k^{(i)} \sim b_k \quad (9)$$

$$\bar{\mathcal{R}}(b_k, \omega_k) \approx \bar{\mathcal{R}}_x(\hat{x}_k, \omega_k) = \bar{\mathcal{R}}_x(\arg \max_x b_k(x), \omega_k) \quad (10)$$

where, $x_k^{(1)}, \dots, x_k^{(n)}$ are n samples drawn from the belief distribution, and $\hat{x}_k$ is the most likely state under b_k .

The definition of the reward $\bar{\mathcal{R}}_x(x_k, \omega_k)$ depends on the selected mapping technique. While the framework can use different measures and mapping methods, in this work we focus on a measure that suits the SPC mapping method. Translating all the SPC mapping performance indices into a single quantitative measure is a challenge. Focusing on information gain and coverage, the measure is derived from mapping requirements in terms of emission, incidence, solar azimuth, and spacecraft azimuth angles. We detail this derivation in the following paragraphs. We start by defining concepts for photometry purposes.

Angles Definition

The definition of three angles is necessary to describe the reflectance properties of surface: incidence angle ι ,

emission angle e , and phase angle ϕ . As it is seen in Figure 6, ι defines the angle between sun incidence and surface normal $\hat{n}$. Emission angle e is the angle between emission ray and surface normal. Phase angle ϕ is the angle between the vector connecting the sun to the center of the body and the vector connecting the spacecraft to the center of the body. Moreover, solar and spacecraft azimuth angles can be defined and they will be used as indicators of the goodness of the mapping phase. The solar azimuth angle (α) is the angle between local north and the projection of the vector from the sun onto the local surface. The spacecraft azimuth angle (β) is the angle between local north and the projection of the vector from the surface to the spacecraft onto the local surface.

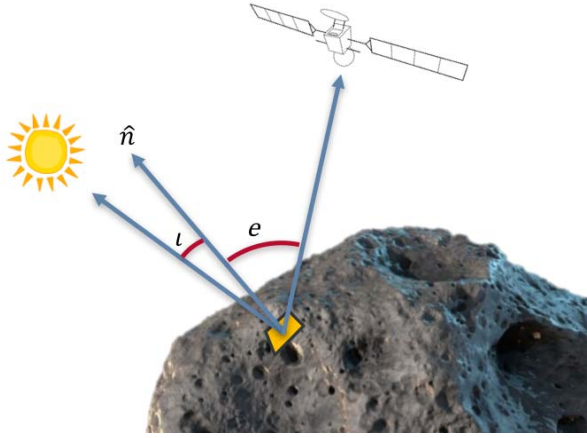


Figure 6. Angles definition

Emission Angle Requirements

In small body mapping, the core of the algorithm is a least square solver. The variation in emission angle increases the robustness of the solver since it minimizes the weighted sum of square-residuals between the extracted (from real images) and predicted brightness (with rendered images). In fact, the brightness of a point on the surface is dependent on the emission and incidence angles. In particular, large variation in emission angle is desirable because, in this case, the orbit offers a great variety of observation geometry of the same portion of the body. Beneficial conditions are achieved when images of each facet are taken at, at least, three distinct emission angle bins in intervals 20-30°, 30-40°, 40-50°, and 50-60°. Moreover, at least one image should have an emission angle less than 10°. This last condition is particularly advantageous to acquire pictures of the limb of the body. The limb represents a known pointing condition and therefore is an additional constraint in the spacecraft state estimation process. Mathematically, the

above-mentioned requirements can be translated into

$$S_{e,k}^i = f_{e,k}^i + l_{e,k}^i \quad (11)$$

where

$$f_{e,k}^i = U\left(\sum_{s=2}^5 U(e_k^i, 10s, 10s + 10), 3, \infty\right) \quad (12)$$

Function $U(e, a, b)$ returns one if $a \leq e \leq b$ and otherwise returns zero.

$$l_{e,k}^i = U(e_k^i, 0^\circ, 10^\circ). \quad (13)$$

Incidence Angle Requirements

The incidence angle is an indicator of the illumination condition of the body. In general, for SPC, the best condition for incidence angle is when images are taken with an angle between 20° and 60°. The reason is that the small angles result in images with excessive brightness that can saturate the image and large angles produce images with excessive shadows and dark areas. Mathematically, the desired score is represented as

$$S_{\iota,k}^i = U(\iota_k^i, 20^\circ, 60^\circ) \quad (14)$$

Solar Azimuth Angle Requirements

The solar azimuth angle is another indicator of the variation of the illumination conditions. As previously explained, for SPC, it is advantageous to have a great variety of illumination levels. In other words, for a given orbit, it is desirable that the maximum variation of the solar azimuth is greater than 120°. This can be expressed as

$$S_{\alpha,k}^i = U(|\max(\alpha_k^i) - \min(\alpha_k^i)|, 120^\circ, 360^\circ) \quad (15)$$

Spacecraft Azimuth Angle Requirements

On the contrary, the spacecraft azimuth angle indicates how the spacecraft position changes with respect to the small body. It is desirable to ensure robust image matching across different views of the same facet. Therefore, we limit the maximum variation in the viewing angle to 100°. Therefore

$$S_{\beta,k}^i = U(|\max(\beta_k^i) - \min(\beta_k^i)|, 0^\circ, 100^\circ) \quad (16)$$

Overall Single-facet Score Function

The overall score function for the i -th facet is the combination of the previous contributions:

$$S_k^i = S_{e,k}^i + S_{\iota,k}^i + S_{\alpha,k}^i + S_{\beta,k}^i \quad (17)$$

Total Reward

The score function defined in equation 17 captures the value of information in the observation. One needs to quantify how much this information contributes to the map information. For example, if a facet has already been observed many times in favorable angles, there will be limited gain in observing it again. In general, the gain in map accuracy for a facet depends on the previous observations from that facet. The information stored in the i -th facet can be expressed as

$$B_k^i = \sum_{k=0}^T S_k^i \quad (18)$$

The contribution of the measured score S_k^i (of the i -th facet at the k -th time step) is weighted by $w_k^i = \xi(|\xi + \beta L_k^i|)^{-1}$ where $L_k^i = 1 - e^{-\psi B_k^i}$ is a quantity bounded between 0 and 1, representing the map accuracy. β , ξ and ψ are tuning parameters. The trend of the weight, for different values of the tuning parameter ξ and for a fixed $\beta = 500$ is shown in Figure 7. Figure 7 confirms that the facets that have already been observed well are not affected much by new observations, and the facets that are poorly observed are highly influenced by addition of the new information. Finally, the reward function can be defined as the information gain at every time step for every facet:

$$\bar{\mathcal{R}}_k = \sum_{i=1}^n w_k^i S_k^i \quad (19)$$

Therefore, the corresponding planning problem is

$$\bar{\pi}^* = \arg \max_{\bar{\pi}} \sum_{k=0}^T \bar{\mathcal{R}}_k(x_k, \omega_k) \quad (20)$$

This results in selecting the orbit that maximizes the information gain of the poorly observed regions thereby increasing their map accuracy. Finally, we can define the reward as

7. RESULTS

In this section we present numerical results of the proposed algorithm. In our simulations to create a high-fidelity model of the small body shape and gravity we rely on JPL's Small Body Dynamics Toolkit (SBDT)[19] and the Primitive Body Coverage and Geometry Evaluator (PB-CAGE) library [20].

For this simulation, we adopt a model of the comet CG67-P where its surface is represented by a mesh with 1146 vertices and 2288 facets. For a given orbit, we evaluate the reward, introduced in Section 5, for each

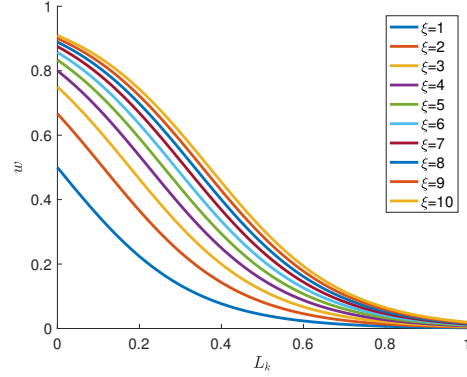


Figure 7. Weight function trend

facet while taking into account the position of spacecraft, small body, and the Sun. We evaluate the score for a set of circular orbits of radius 10 km, centered in the small body. Each orbit is characterized by a different inclination angle ranging from 0° to 170° with a delta-angle of 10° .

We compare the proposed method for autonomous selection of optimal orbit sequence (OOS) with two baseline approaches: fixed inclination orbit sequence (FIOS) and random sequence of orbits (RSO). In the FIOS baseline, the user selects a suitable orbit at the beginning of the mapping and the orbit does not change over time. In our simulation, we selected a polar orbit for the FIOS case. In RSO, at each step, we pick a random orbit from the set of candidate orbits.

The performance and comparison metric among these three different alternatives is the overall coverage score. A higher score is translated into a better coverage, resulting in a better map. In fact, acquiring good images is mandatory for the correct behavior of the SPC mapping algorithm. Bad images can lead the process to diverge and, therefore, implying heavier human effort. Designing a good trajectory is the first step to making the mapping procedure easier and, in principle, automatic.

A sequence of three orbits is considered for each case. Figure 8 shows the evolution in time of the maximum score for each facet.

The yellow areas represent the part of the surface that have an higher coverage score after one orbit. As can be seen in Figure 8a, the OOS guarantees an increase of the coverage score in time. In Figure 8b, RSO shows an overall lower score compared to OOS, with some poor areas. Finally, in Figure 8c, as expected, the coverage score for the FIOS case is constant in time. To have a more quantitative idea of the results, Table 2 reports the total coverage score at the end of the third orbit. This value is computed by summing the maximum score for each facet at the end of the mapping phase.

Moreover, Table ?? and Figure 9a report the number of

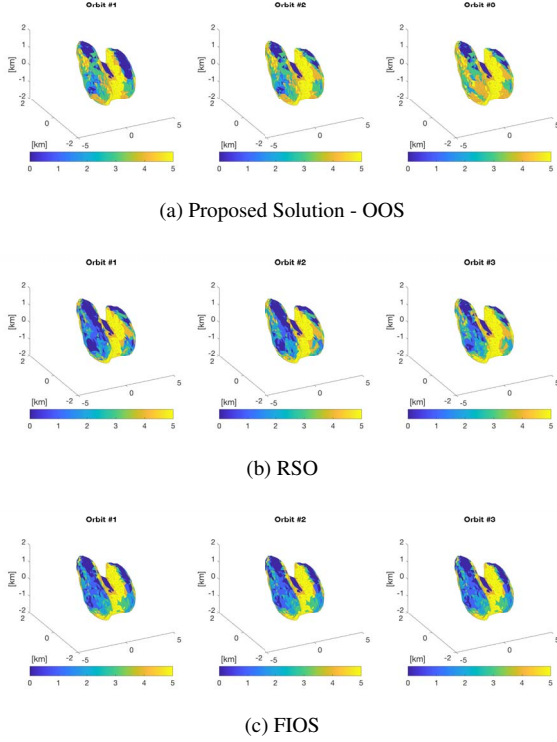


Figure 8. Maximum facet score evolution

Table 2. Total coverage score

Strategy	Total Coverage Score
Proposed Solution - OOS	8956
RSO	7992
FIOS	6875

facets with global score < 3 at the end of the mapping phase. This is an indication of the number of poor covered regions at the end of the mapping phase.

Table 3. Poor coverage regions

Strategy	Poor Score Facets
Proposed Solution - OOS	283
RSO	576
FIOS	867

Figure 9a shows how the selection of the optimal sequence offers a significant improvement in the global coverage of the body. Finally, Figure 9b shows the time evolution of the facets with score equal to 5, the ideal condition in terms of coverage. According to the measures in Figure 9, OOS supersedes other methods in the number of facets with maximum score.

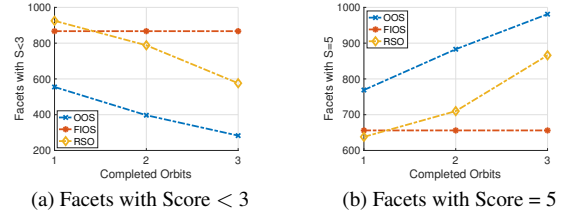


Figure 9. Low and highest score facets

8. CONCLUSION

In this work, we have proposed an algorithm for autonomous orbit selection and adaptation around small bodies while mapping its surface. Focusing on the stereophotoclinometry (SPC) mapping method, we developed cost functions that quantify the orbit goodness in the sense of map improvement. These mapping accuracy measures are then used to guide the orbit selection process. We relied on a reduced policy space where orbits are selected from a representative family of common orbits around small bodies. Via numerical simulations, we showed that the optimal sequence of orbits, resulting from the optimization process, offers a substantial improvement of the mapping performance compared to the fixed orbit case. The proposed algorithm is a step toward a higher level of autonomy and on-board trajectory planning around small bodies and reduces the human involvement in this process. As a future work, we will incorporate the small body gravity and high-fidelity orbital dynamics into the formulation.

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